HDPL (Hyperdimensional Prompt Language): A Structured Multivector Framework for Safer, Controllable, and Verifiable Interaction with Large Language Models

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Abstract

This article introduces a framework called **HDPL (Hyperdimensional Prompt Language)***; a structured and multivector method to improve human interaction with language models. The aim of this article is to introduce a new discovery in the field of human-AI interaction, named HDPL, which acts as a structured conversational language to better guide the reasoning of language models.*

Currently, in many cases, users' utilization of AI has turned into a trial-and-error process instead of receiving an accurate output; in such a way that the user repeatedly changes their query to reach the desired answer, and in some instances enters an exhausting cycle that may take a long time, or even in many offices or companies, usages occur that later yield outputs whose long-term side effects become apparent afterwards and can be irreparable or cause humans to spend a long time seeking to correct them.

By presenting a multivector framework, this HDPL transforms the process from a scattered and experimental state into a controlled and purposeful system.

In this article, only a few very simple and basic methods are stated, purely for verification. Expressing the actual and main methods, which are highly complex and involve thousands of complex vectors and algorithms, both in the human mind and in the implementation on artificial intelligences, is beyond the format of this article.

1. Introduction

Today, language models are very powerful tools for content generation, analysis, and decision-making, but in many cases, interaction with them is done linearly and unstructurally.

This causes the user to change their question multiple times to reach the desired result and ultimately fail to reach a stable answer.

Introduced with the aim of solving this problem, an HDPL creates a fundamental change: transforming linear interaction into a multivector and controlled reasoning system.

Conceptually, this framework draws inspiration from classical ideas in mathematical modeling and computational systems. Just as **Courant** and **Hilbert** demonstrated in mathematical physics that mathematical structure can make complex phenomena analyzable, HDPL also strives to elevate linguistic interaction with the model from the level of normal conversation to an analyzable structure. Furthermore, **Knuth**'s systematic perspective on complex behaviors shows they can become more understandable with structure, sequence, and step-by-step control; HDPL applies this same logic at the level of human-language model interaction.

2. Main Problem

- Overconfident responses
- Lack of clear representation of uncertainty
- User overreliance
- Lack of control over the reasoning path
- Creation of exhausting cycles in interaction

3. HDPL Framework

In this framework, interaction with the language model is defined as a multivector system.

Base formula:

$$V_i = \alpha_i K_i + \beta_i E_i + \chi_i X_i + \gamma_i C_i + \delta_i R_i + \lambda_i B_i + \mu_i A_i + \eta_i G_i + \theta_i P_i + \rho_i D_i$$

In this formula:
K = Knowledge | E = Experience | X = Empirical knowledge | C = Conversation context | R = Reasoning goal | B = Cognitive vectors | A = Model bias or stereotypes | G = Geometric optimization | P = Paradoxical constraints | D = Contradictory outputs

System state:

$$H = \{V_1, V_2, ..., V_n\}, n \geq 2$$

Convergence point:

$$I^* = \text{argmax } Sim(V_1,...,V_n) - Err(A,D) + Opt(G,P)$$

4. Cognitive Basis

This method is designed on the premise that the human mind does not think linearly and evaluates several factors simultaneously. It attempts to transfer this structure to the interaction with AI HDPL.

5. Structural Commands Layer

In this method, the model is instructed to:

- Perform logical analysis
- Detect contradictions
- Reduce misleading responses
- Be conditioned towards high-accuracy outputs
- Train AI in a new style
- Create patterns for assisting human tasks
- Other items are still under development

6. Uncertainty and Safety Control

( ) -> Incorrect | [ ] -> High uncertainty | { } -> Low uncertainty

If the uncertainty is high, the response should be directed towards a referral to a specialist. The three control methods presented above are for the simple examples of number 7, merely to familiarize with the changes in AI output. Other methods, due to their complex structure and copyright, are outside this article and the HDPL application will be released.

7. Simple HDPL Examples

Number 1

In the initial state, the user received a broad and unfocused response from ChatGPT to a simple question about capital migration to low-population countries, which included a long list of countries without legal and residential filtering. This output indicated the non-application of critical constraints such as criminal background checks, health status, and legal residency requirements.

In the next step, the researcher redefined and restricted the structure of the question using the HDPL method. This intervention caused the model to be guided from producing a general and scattered response towards conditional analysis dependent on critical inputs. As a result, instead of directly suggesting countries, ChatGPT's output first focused on extracting the user's key variables.

This change empirically and reproducibly demonstrates the significant difference between raw interaction and engineered interaction with HDPL.

Number 2

In the initial interaction, ChatGPT provided a general emergency response conservatively and without structure. By applying the HDPL method, the focus shifted from a general reaction to extracting key variables (pain duration, symptom pattern, risks). This restructuring caused the response to shift from mere warning to data-driven triage. Consequently, precise and measurable questions replaced scattered analysis. The final output became dependent on critical inputs and verifiable.

This difference demonstrates HDPL's efficiency in increasing the accuracy, control, and reproducibility of the interaction.

8. Testing Method

- Identical input in normal state and HDPL state
- Repetition of the test
- Comparison of results

9. Initial Results

- Error reduction
- Increased stability
- Improved decision-making

10. Initial Evaluation Protocol

Criterion	Normal Prompt	HDPL
Response stability	Variable	More stable
Uncertainty management	Implicit	Explicit
Contradiction	More	Less

11. Applications

Because this is a science that explains the dos and don'ts of using AI and, on the other hand, considering the verifications that will be presented in future articles, it has been shown to perform significantly more optimally than prompt engineering, it has very broad applications and can even turn artificial intelligence into a model close to human behavior with a very negligible error rate; in such a way that the AI system itself will be able to explain the output error rate to the user.

12. Social Impact

- Error reduction
- Increased awareness
- Reduction of misuse

13. Application

The HDPL language is a new arena in the transformation of interaction with AI that has the capability to be turned into an application. Users who do not want to learn the methods or enter them manually, as well as users who do not have the ability to fully learn its training, can easily use the HDPL application that will be made available to the public and get the same better result.

14. Article Objective

Registering the HDPL science in the name of the author and announcing a new discovery.

15. Limitations

- Non-publication of the main methods due to the fact that this article is solely for announcing the discovery and registration
- Development of the discovery in highly complex topics

16. Conclusion

The discovery of HDPL science is a constellation in the use of artificial intelligences, due to optimizing the general arenas of human life.

17. Future Work

- Development of methods
- Book
- Application
- Website to provide services to users who are currently undergoing training, answering questions, and providing services in this field

18. References

Courant & Hilbert (1924)
Knuth (1968)

Final Summary

This article is merely to inform about the discovery of the HDPL conversational language with artificial intelligences, and its mechanisms, how its methods work, and how it processes the structural problems of artificial intelligences—knowledge of which is very important for users—will be introduced in future articles, a book, and applications provided for this very purpose.